From target states to two common grades

One joint law; w = (3/5, 2/5),
a = (3/4, 1/4), τ = 1/5; M = 6.

(a1) State 1: probability 3/4

Unit	θ_1	θ_2	Composite c
A	4	4	4
B	3	2	11/4
C	2	3	9/4
D	1	1	1

(a2) State 2: probability 1/4

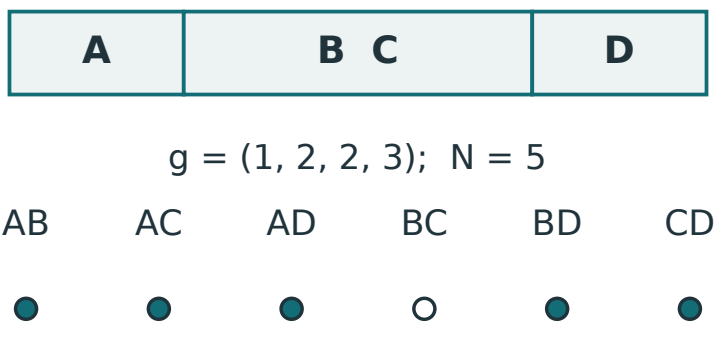
Unit	θ_1	θ_2	Composite c
A	3	1	5/2
B	4	2	7/2
C	2	3	9/4
D	1	4	7/4

$c_i = (3/4)\theta_{i1} + (1/4)\theta_{i2}$
w weights directional errors.
a combines target magnitudes.

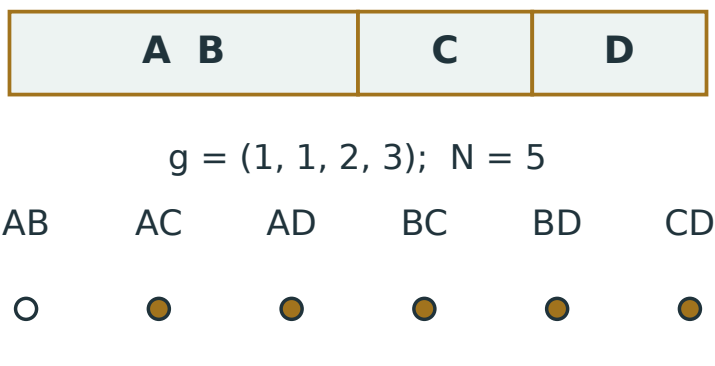
(b) Forward error coefficients

	AB	AC	AD	BC	BD	CD
Weighted direction	1/4	1/10	1/10	2/5	1/10	1/10
Composite	1/4	0	0	0	0	0

(c) Weighted-direction optimum 1223



(c) Composite optimum 1123



Filled: forward direction. Open: withhold.

(d) Same N = 5; evaluate each loss

Report	Weighted risk	Composite risk
1223	**17/120**	3/40
1123	1/6	**1/30**

Risk uses M = 6; fractions use N.
Lower losses across targets do not choose the substantively appropriate target.